

Dipayan Biswas

Applied Scientist — Machine Learning Engineer — Multimodal AI, Computer Vision

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SUMMARY

PhD Candidate in Computer Science with 5+ years of experience in computer vision and multimodal AI. Expertise in developing end-to-end ML pipelines for video understanding and multimodal summarization, with automated evaluation frameworks using PyTorch and transformer-based architectures.

EDUCATION

University of Houston

Ph.D. in Computer Science | GPA: 3.88

Houston, TX

Aug 2021 – Jul 2026 (Expected)

Khulna University of Engineering and Technology (KUET)

B.Sc. in Electronics and Communication Engineering | GPA: 3.88 | University Gold Medalist

Khulna, Bangladesh

Apr 2012 – May 2016

EXPERIENCE

Fellowship.AI

Machine Learning Fellow

Remote

Sept 2025 – Nov 2025

- Engineered a real estate data pipeline to scrape listings and perform nearest-neighbor comparative market analysis on a 4.5GB Redfin dataset, estimating property values across localized markets.
- Built a Gemini-based multimodal workflow to deliver structured recommendations for pricing and content optimization by analyzing listing images and descriptions.

University of Houston

Graduate Research Assistant

Houston, TX

Aug 2021 – Present

- Advanced **multimodal AI methods** for educational video understanding, focusing on multimodal summarization that integrates text descriptions with representative supporting visuals.
- Integrate multi-channel informations** including ASR transcripts, OCR text, slide visuals, and visual object detection in multimodal pipelines for chapter-level educational video summarization.
- Built **automated LLM-based evaluation workflows** with DeepEval and ChatGPT/Gemini APIs to assess summaries, improving factual faithfulness by 9.6 points and relevance by 5.2 points over text-only baselines.
- Improved visual content detection in educational videos by optimizing YOLOv11 with transfer learning and semi-supervised auto-labeling, increasing AP50 from 90.75% to 95.32%.
- Developed LogiForm, a graph-based computer vision algorithm using color and keypoint similarity to detect meaningful visual objects in lecture videos, improving mAP by 15.8%.

Khulna University of Engineering and Technology (KUET)

Assistant Professor / Lecturer

Khulna, Bangladesh

Feb 2017 – Aug 2021

- Developed and benchmarked a ResNet50 transfer-learning model for plant disease detection, achieving 99.80% accuracy and outperforming prior deep learning baselines.
- Built deep learning and classical ML models for medical diagnosis, including chest X-ray pneumonia detection and breast cancer prediction, achieving up to 99.28% accuracy.

Samsung R&D Institute Bangladesh (SRBD)

Software Engineer

Dhaka, Bangladesh

Jul 2016 – Feb 2017

- Developed 360-degree image rendering features, including panoramic, 360-view, and little-planet visualizations, for the Samsung Gear 360 iPhone application using Objective-C.
- Optimized mobile image processing and rendering pipelines with OpenCV and OpenGL, improving visual quality and reducing latency in resource-constrained environments.

PROJECTS

Multimodal Video Summarization Engine

Dec 2025 - Mar 2026

- Built an end-to-end multimodal summarization system for the NSF-supported VideoPoints platform to improve educational video navigation and review.
- Engineered multimodal fusion workflows using Whisper, Gemma 3/Qwen3-VL, and Gemini Flash with slide-structured in-context learning to align slide-level evidence and improve synthesis quality.

LLM-Based Video Temporal Segmentation

June 2025 - Aug 2025

- Built a slide-anchored LLM-based video chaptering pipeline that aligned Whisper transcript segments with slide transitions to partition long educational videos into topically coherent chapters.
- Applied validation, retry, and correction logic to ensure contiguous chapter boundaries and full video coverage.

SOTA Object Detection Lab | [Code](#)

Dec 2024 - Feb 2025

- Built a PyTorch-based framework for training and fine-tuning state-of-the-art object detectors on custom datasets.
- Developed reusable experiment infrastructure with Hydra-based configuration, dynamic model loading, and shared train/evaluate/predict workflows to streamline reproducible ML experiments.

Long-Distance Face Detection | [Code](#)

Spring 2023

- Enhanced RetinaFace with an IoU-aware multi-task loss and auxiliary IoU head for long-distance face detection.
- Trained and evaluated ResNet50 and MobileNet0.25 backbones on WIDER FACE, improving robustness for detecting small, distant, and occluded faces.

SELECTED PUBLICATIONS ([GOOGLE SCHOLAR](#))

D. Biswas, S. Shah, and J. Subhlok. “Visual Content Detection in Educational Videos with Transfer Learning and Dataset Enrichment”. In *IEEE MIPR*, 2025. [[Paper](#)] [[Code](#)]

D. Biswas, S. Shah, and J. Subhlok. “Lecture Video Visual Objects (LVVO) Dataset: A Benchmark for Visual Object Detection”. In *arXiv preprint arXiv:2506.13657*, 2025. [[Paper](#)] [[Dataset](#)]

D. Biswas, S. Shah, and J. Subhlok. “Identification of Visual Objects in Lecture Videos with Color and Keypoints Analysis”. In *IEEE ISM*, 2023. [[Paper](#)]

SKILLS

Languages: Python, SQL, C++, MATLAB, R, PHP, JavaScript

Frameworks/Libraries: PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, DeepEval, Hydra, Lightning

Tools/Platforms: Git, Docker, Linux, Slurm, MLflow, W&B, MySQL, AWS EC2, AWS SageMaker, LangChain

HONORS AND AWARDS

Ph.D. Research Showcase Award (Audience’s Choice) — University of Houston, 2025

Cullen Graduate Student Fellowship — Graduate School, University of Houston, 2021-2026

University Gold Medalist — Khulna University of Engineering & Technology (KUET), Bangladesh, 2018

LEADERSHIP

Team Lead, Real Estate Listing Optimization Project, Fellowship.AI

Sep 2025 - Nov 2025

Organizing Secretary, Graduate Student Association of Bangladesh, University of Houston

Aug 2022 - Jul 2025

Academic Advisor and Project Mentor, Khulna University of Engineering and Technology

Feb 2017 - Aug 2021

CERTIFICATIONS

[Generative AI with Large Language Models](#), AWS and DeepLearning.AI (Coursera)

[Machine Learning in Production](#), DeepLearning.AI (Coursera)

[AWS Cloud Practitioner Essentials](#), Amazon Web Services (AWS)

[Deep Learning Specialization](#), DeepLearning.AI (Coursera)

REFERENCES

[Shishir Shah](#), Chief AI Officer & Director of School of Computer Science, University of Oklahoma Email: sshah@ou.edu
[Jaspal Subhlok](#), Professor, Department of Computer Science, University of Houston Email: jsubhlok@central.uh.edu